

RELATORIO TAF x METAR - grupo CIMAER

Gerado 2026-09-01 11:15 | 87 aerodromos | fonte: excel/malha.csv + tabelas metars / tafs

Secoes: 1) presenca por estacao x periodo 2) fenomenos perigosos, acuracia 3) perigosos por estacao x periodo 4) consolidado. Matrices por aeroporto em output/*_CIMAER.csv.

TAF REDEMET vs METAR - Estacao x Periodo | grupo CIMAER

Gerado 2026-09-01 11:15 | malha.csv | 134,066 voos comparaveis | 33 aeroportos com dado

Regra: acerto = TAF e METAR concordam na presenca de fenomeno (ambos com condicao, ou ambos NSW). Falso positivo = TAF previu, METAR nao confirmou. Miss = METAR teve condicao, TAF nao previu.
Janela: 'Hora Chegada Alternado' UTC. Estacoes astronomicas (hem. sul).

Geral

Comparaveis	Certo	%	Falso Positivo	Miss
134,066	109,493	81.7%	14,227	10,346

Por estacao

estacao	Comparaveis	Certo	%	Falso Pos.	Miss
Verao	32,069	25,019	78.0%	3,708	3,342
Outono	33,684	27,357	81.2%	3,643	2,684
Inverno	34,994	30,046	85.9%	3,222	1,726
Primavera	33,319	27,071	81.2%	3,654	2,594

Por periodo (UTC)

(UTC)	Comparaveis	Certo	%	Falso Pos.	Miss
00-06	24,699	19,939	80.7%	2,640	2,120
06-12	21,078	15,226	72.2%	3,990	1,862
12-18	44,168	38,639	87.5%	2,482	3,047
18-24	44,121	35,689	80.9%	5,115	3,317

Estacao x periodo (% acerto / n)

Estacao \ Periodo	00-06	06-12	12-18	18-24	TODOS
Verao	75.2% (5,920)	78.2% (5,103)	85.2% (10,544)	72.3% (10,502)	78.0% (32,069)
Outono	83.0% (6,265)	62.6% (5,090)	86.9% (11,237)	83.0% (11,092)	81.2% (33,684)
Inverno	84.3% (6,329)	73.2% (5,454)	90.4% (11,468)	88.1% (11,743)	85.9% (34,994)
Primavera	80.1% (6,185)	74.7% (5,431)	87.1% (10,919)	79.2% (10,784)	81.2% (33,319)
TODAS	80.7% (24,699)	72.2% (21,078)	87.5% (44,168)	80.9% (44,121)	81.7% (134,066)

Aeroportos | grupo CIMAER

Gerado 2026-09-01 11:15 | 33 aeroportos com voo comparavel | ordenado por volume

ICAO	Comparaveis	Certo	%	Falso Pos.	Miss
SBGL	39,347	31,251	79.4%	5,251	2,845
SBCT	27,384	21,289	77.7%	3,578	2,517
SBFL	10,095	8,592	85.1%	907	596
SBIP	7,122	6,203	87.1%	589	330
SBCF	6,087	5,235	86.0%	373	479
SBJU	4,427	4,133	93.4%	196	98
SBBR	4,187	3,632	86.7%	250	305
SBRF	4,129	3,296	79.8%	428	405
SBBV	3,750	3,041	81.1%	377	332
SBRD	2,914	2,436	83.6%	201	277
SBSV	2,767	2,157	78.0%	235	375
SBAU	2,708	2,476	91.4%	129	103
SBMG	2,317	1,997	86.2%	109	211
SBIH	1,998	1,630	81.6%	225	143
SBMO	1,723	1,428	82.9%	123	172
SBCG	1,601	1,387	86.6%	73	141
SBFI	1,412	1,197	84.8%	104	111
SBSG	1,321	1,054	79.8%	96	171
SBSL	1,106	890	80.5%	162	54
SBSM	1,089	928	85.2%	86	75
SBPA	1,069	890	83.3%	69	110
SBCA	1,042	883	84.7%	64	95
SBBE	946	713	75.4%	114	119
SBSR	679	629	92.6%	25	25
SBAT	637	518	81.3%	57	62
SBCY	572	427	74.7%	118	27
SSKW	493	398	80.7%	35	60
SBEG	439	278	63.3%	115	46
SBPV	363	265	73.0%	55	43
SBRB	223	139	62.3%	70	14
SBFZ	104	90	86.5%	12	2
SBZM	14	10	71.4%	1	3
SBDN	1	1	100.0%	0	0

Acuracia - fenomenos perigosos | grupo CIMAER

Gerado 2026-09-01 11:15 | malha.csv | 134,066 voos comparaveis | TAF REDEMET vs METAR observado

Lista: FZDZ FZRA FZFG TS +TS +RA +SHRA +SN +SHSN TSRA. 'QUALQUER perigosa' = o voo tem pelo menos um desses tokens. Acuracia = acerto / (acerto + falso positivo + miss).

Leitura	Voos c/ perigo	Acerto	Falso pos.	Miss	Acuracia	Deteccao	Precisao
exato	7,606	459	6,296	851	6.0%	35.0%	6.8%
sem intensidade	8,417	821	6,172	1,424	9.8%	36.6%	11.7%

Deteccao = acerto / observados (1 - miss). **Precisao** = acerto / sinalizados (1 - falso positivo).

Por fenomeno

Gerado 2026-09-01 11:15 | 134,066 voos comparaveis

Visao EXATA

Fenomeno	TAF previu	Confirmado	Precisao %	METAR teve	Detectado %	Falso pos.	Miss
TS	1,450	62	4.3%	830	7.5%	1,388	768
+RA	22	0	0.0%	46	0.0%	22	46
+SHRA	0	0	-	6	0.0%	0	6
+TSRA	72	0	0.0%	96	0.0%	72	96
TSRA	5,211	134	2.6%	334	40.1%	5,077	200

Visao SEM INTENSIDADE

Fenomeno	TAF previu	Confirmado	Precisao %	METAR teve	Detectado %	Falso pos.	Miss
FZDZ	0	0	-	0	-	0	0
FZFG	0	0	-	0	-	0	0
FZRA	0	0	-	0	-	0	0
TS	1,450	62	4.3%	830	7.5%	1,388	768
TSRA	5,521	499	9.0%	1,365	36.6%	5,022	866
+RA	22	0	0.0%	46	0.0%	22	46
+SHRA	0	0	-	6	0.0%	0	6
+SHSN	0	0	-	0	-	0	0
+SN	0	0	-	0	-	0	0

Condicoes perigosas - TAF REDEMET vs METAR | grupo CIMAER

Gerado 2026-09-01 11:15 | malha.csv | 134,066 voos comparaveis | 33 aeroportos

Acerto TAF % = dos voos em que o TAF previu a condicao, quantos o METAR confirmou (match exato).
Detectado % = dos voos com a condicao no METAR, quantos o TAF previra.
Congelantes/+TS/+SHRA/+SN/+SHSN omitidos (zero no TAF). Estacoes astronomicas (hem. sul), periodos UTC.

Visao EXATA (token identico) - geral

Cond	TAF previu	Confirmado	Acerto TAF %	METAR teve	Detectado %
TS	1,450	62	4.3%	830	7.5%
+RA	22	0	0.0%	46	0.0%
+TSRA	72	0	0.0%	96	0.0%
TSRA	5,211	134	2.6%	334	40.1%

Visao SEM INTENSIDADE (+/- removido) - geral

Cond	TAF previu	Confirmado	Acerto TAF %	METAR teve	Detectado %
TS	1,450	62	4.3%	830	7.5%
TSRA	5,521	499	9.0%	1,365	36.6%
+RA	22	0	0.0%	46	0.0%

Por estacao - visao exata

estacao \ cond	TS	+RA	+TSRA	TSRA
Verao	6.3% (527)	-	0.0% (30)	1.6% (2,351)
Outono	1.2% (256)	0.0% (22)	-	2.3% (946)
Inverno	3.0% (165)	-	0.0% (20)	5.1% (509)
Primavera	4.2% (502)	-	0.0% (22)	3.4% (1,405)

Por periodo (UTC) - visao exata

periodo \ cond	TS	+RA	+TSRA	TSRA
00-06	2.6% (151)	-	-	5.4% (569)
06-12	0.0% (82)	-	-	2.2% (325)
12-18	2.6% (154)	-	0.0% (22)	3.1% (947)
18-24	5.1% (1,063)	0.0% (22)	0.0% (50)	2.0% (3,370)

TSRA - estacao x periodo (acerto TAF % / n previsto)

Estacao \ Periodo	00-06	06-12	12-18	18-24	TODOS
Verao	1.5% (268)	1.8% (110)	0.7% (306)	1.8% (1,667)	1.6% (2,351)
Outono	2.9% (105)	4.3% (47)	3.8% (157)	1.7% (637)	2.3% (946)
Inverno	0.0% (46)	0.0% (45)	7.6% (132)	5.6% (286)	5.1% (509)
Primavera	16.0% (150)	2.4% (123)	3.1% (352)	1.3% (780)	3.4% (1,405)
TODAS	5.4% (569)	2.2% (325)	3.1% (947)	2.0% (3,370)	2.6% (5,211)

TS - estacao x periodo (acerto TAF % / n previsto)

Estacao \ Periodo	00-06	06-12	12-18	18-24	TODOS
Verao	3.0% (66)	0.0% (6)	3.9% (51)	7.2% (404)	6.3% (527)
Outono	4.2% (24)	0.0% (22)	0.0% (40)	1.2% (170)	1.2% (256)
Inverno	0.0% (11)	0.0% (18)	0.0% (13)	4.1% (123)	3.0% (165)
Primavera	2.0% (50)	0.0% (36)	4.0% (50)	4.9% (366)	4.2% (502)
TODAS	2.6% (151)	0.0% (82)	2.6% (154)	5.1% (1,063)	4.3% (1,450)

Aeroportos - TS/TSRA (visao exata) | grupo CIMAER

Gerado 2026-09-01 11:15 | 31 aeroportos com TS/TSRA no TAF ou METAR. Ordenado por nº previsto.

ICAO	Voos	TAF previu	Confirmado	Acerto %	METAR teve	Detectado %
SBGL	39,347	1,636	90	5.5%	217	41.5%
SBCT	27,384	1,161	86	7.4%	268	32.1%
SBCF	6,087	467	33	7.1%	78	42.3%
SBBV	3,750	409	11	2.7%	52	21.2%
SBBR	4,187	307	22	7.2%	71	31.0%
SBIP	7,122	300	6	2.0%	11	54.5%
SBFL	10,095	263	12	4.6%	39	30.8%
SBRD	2,914	251	5	2.0%	28	17.9%
SBIH	1,998	200	16	8.0%	85	18.8%
SBEG	439	183	16	8.7%	19	84.2%
SBSL	1,106	146	6	4.1%	11	54.5%
SBMG	2,317	143	3	2.1%	27	11.1%
SBCY	572	137	5	3.6%	12	41.7%
SBJU	4,427	134	9	6.7%	28	32.1%
SBCG	1,601	118	11	9.3%	38	28.9%
SBAU	2,708	111	17	15.3%	64	26.6%
SBBE	946	107	3	2.8%	18	16.7%
SBFI	1,412	107	11	10.3%	20	55.0%
SBSM	1,089	78	8	10.3%	14	57.1%
SBAT	637	63	2	3.2%	7	28.6%
SBCA	1,042	59	6	10.2%	22	27.3%
SBPA	1,069	54	8	14.8%	10	80.0%
SBPV	363	48	6	12.5%	6	100.0%
SBSV	2,767	45	0	0.0%	1	0.0%
SSKW	493	44	2	4.5%	13	15.4%
SBRF	4,129	33	0	0.0%	2	0.0%
SBSG	1,321	21	0	0.0%	0	-
SBRB	223	12	0	0.0%	1	0.0%
SBSR	679	12	0	0.0%	2	0.0%
SBMO	1,723	7	0	0.0%	0	-
SBFZ	104	5	0	0.0%	0	-

TAF REDEMET vs METAR | grupo CIMAER

Gerado 2026-09-01 11:15 | comparaveis: 134,066 de 137,588

Regra: acerto quando TAF e METAR concordam na presenca de fenomeno (ambos com condicao, ou ambos NSW).

	Certo	%	Falso Positivo	Miss
TAF REDEMET	109,493	81.7%	14,227	10,346

TAF REDEMET vs METAR - Condições Perigosas (match exato)

Gerado 2026-09-01 11:15 | comparáveis: 134,066

Condição	TAF previu	METAR igual	Errado	Acerto %
-FZDZ	0	0	0	0.0%
FZDZ	0	0	0	0.0%
+FZDZ	0	0	0	0.0%
-FZRA	0	0	0	0.0%
FZRA	0	0	0	0.0%
+FZRA	0	0	0	0.0%
FZFG	0	0	0	0.0%
TS	1,450	62	1,388	4.3%
+TS	0	0	0	0.0%
+RA	22	0	22	0.0%
+SHRA	0	0	0	0.0%
+SN	0	0	0	0.0%
+SHSN	0	0	0	0.0%
+TSRA	72	0	72	0.0%
TSRA	5,211	134	5,077	2.6%
TOTAL	6,755	196	6,559	2.9%

O que o METAR mostrou, quando o TAF previu cada condição:

TS (previsto 1,450x)

METAR real	Qtd	%
NSW	1,049	72.3%
VCSH	137	9.4%
-RA	61	4.2%
VCTS	48	3.3%
TS VCSH	39	2.7%
-TSRA	31	2.1%
VCSH VCTS	19	1.3%
TS	17	1.2%
Other	49	3.4%

+RA (previsto 22x)

METAR real	Qtd	%
-RA	14	63.6%

-RA BR	8	36.4%
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+TSRA (previsto 72x)

METAR real	Qtd	%
NSW	33	45.8%
-RA	14	19.4%
TSRA	7	9.7%
TS	6	8.3%
VCSH	5	6.9%
-TSRA	5	6.9%
TSRA BR	1	1.4%
TS VCSH	1	1.4%

TSRA (previsto 5,211x)

METAR real	Qtd	%
NSW	3,155	60.5%
-RA	347	6.7%
VCSH	310	5.9%
-TSRA	261	5.0%
VCTS	199	3.8%
-RA BR	155	3.0%
TS VCSH	118	2.3%
TSRA	112	2.1%
Other	554	10.6%

TAF REDEMET vs METAR - Perigosas (intensidade ignorada)

Gerado 2026-09-01 11:15 | comparaveis: 134,066

Condicao base	TAF previu	METAR igual	Errado	Acerto %
FZDZ	0	0	0	0.0%
FZFG	0	0	0	0.0%
FZRA	0	0	0	0.0%
TS	1,450	62	1,388	4.3%
TSRA	5,521	499	5,022	9.0%
+RA	22	0	22	0.0%
+SHRA	0	0	0	0.0%
+SHSN	0	0	0	0.0%
+SN	0	0	0	0.0%
TOTAL	6,993	561	6,432	8.0%